KOMPALWAR GANGOTRI DATTA

B.Tech. - Electronics and Telecommunication Engineering | AI / ML Specialist

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PROFESSIONAL SUMMARY

Final-year B.Tech. Electronics and Telecommunication Engineering student specializing in AI/ML, with hands-on experience designing and developing intelligent applications across Computer Vision, Generative AI, Retrieval-Augmented Generation (RAG), and Voice AI. Strong technical proficiency in Python, PyTorch, TensorFlow, OpenCV, YOLOv8, FastAPI, LLMs, embeddings, vector databases, and semantic search. Knowledge of LangChain and LangGraph for building RAG pipelines, tool/function calling, agentic workflows, state management, and multi-step AI execution. Experienced in implementing practical AI solutions involving real-time object detection, document intelligence, identity verification, multilingual AI tutoring, speech-to-text, text-to-speech, and AI backend integration.

KEY EXPERTISE & TECHNICAL SKILLS

AI / ML & Generative AI: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI (LangChain, LangGraph), LLMs (TinyLLaMA, llama.cpp), Semantic Search, Scikit-learn, PyTorch, TensorFlow, Random Forest, Isolation Forest

Computer Vision & Document AI: OpenCV, YOLOv8 (Object Detection), DeepFace (Biometric Verification), Tesseract OCR, Liveness Detection, Image Processing

Vector Search & Voice AI: FAISS, SentenceTransformers, Embeddings, SpeechRecognition, Google STT, Edge TTS, pyttsx3 (STT/TTS)

Programming & Backend: Python, FastAPI, SQLite, REST APIs, Data Preprocessing, Feature Engineering, Model Evaluation

INTERNSHIPS

SmartBridge (SkillWallet) — **Artificial Intelligence and Machine Learning Intern** *Remote | 16 Jun, 2025 – 05 Aug, 2025*

Key Skills: Python, Machine Learning, Data Preprocessing, Feature Engineering, Predictive Analytics

- Completed an experiential internship in Artificial Intelligence and Machine Learning at SmartBridge (SkillWallet), gaining hands-on experience in Python-based machine learning workflows.
- Performed structured data preprocessing, feature engineering, exploratory data analysis, and predictive model development.
- Applied machine learning algorithms and predictive analytics techniques to solve real-world problems, strengthening practical skills in AI/ML model deployment and evaluation.

PROJECTS

GramNet AI: Offline Multilingual RAG-based AI Tutor **[github.com/kompalwargangotri/gramnet-ai-offline-multilingual-rag-tutor]**

Key Skills: RAG, LLMs, TinyLLaMA, Semantic Search, FAISS, SentenceTransformers, Python, SpeechRecognition/Google STT, Edge TTS, pyttsx3, LangChain

- Developed an AI-powered multilingual tutoring system using Retrieval-Augmented Generation (RAG) and TinyLLaMA to deliver context-aware educational responses grounded in retrieved knowledge.
- Implemented a semantic retrieval pipeline using text chunking, SentenceTransformers embeddings, and FAISS for efficient knowledge retrieval and context selection.
- Integrated a locally hosted TinyLLaMA model through llama.cpp for response generation; enabled multilingual voice interaction in English, Hindi, and Marathi via STT/TTS integration with adaptive quizzes optimized for offline-first usage.

SecureFin: YOLO-Based Automated Video KYC and AML Fraud Detection System **[github.com/kompalwargangotri/securefin-video-kyc-aml]**

Key Skills: YOLOv8, Tesseract OCR, DeepFace, TensorFlow, Computer Vision, Object Detection, Image Processing, Isolation Forest, FastAPI, SQLite

- Developed an AI-powered Video KYC and Anti-Money Laundering (AML) fraud detection system for automated identity verification and suspicious activity detection.
- Integrated Tesseract OCR to extract and validate Aadhaar/PAN details, YOLOv8 for person detection and single-person verification, and DeepFace for facial recognition and biometric identity verification.
- Implemented liveness detection to strengthen video-based identity checks, Isolation Forest for identifying suspicious AML activity, and a FastAPI backend with SQLite for structured data management.

NephroPredict: AI-Based Early Detection of Chronic Kidney Disease (CKD) **[github.com/kompalwargangotri/nephropredict-ai-ckd-detection]**

Key Skills: Python, Machine Learning, Scikit-learn, Pandas, NumPy, Data Preprocessing, Feature Engineering, Model Evaluation, Data Visualization

- Developed a machine learning-based system for early prediction of Chronic Kidney Disease (CKD) using clinical patient healthcare data.
- Performed data preprocessing and feature engineering using Pandas and NumPy; trained a Random Forest classifier using Scikit-learn, achieving 90.96% accuracy and high F1-score for real-time risk prediction.

EDUCATION

MIT Academy of Engineering, Pune — B.Tech. in Electronics and Telecommunication Engineering | CGPA: 7.90 / 10 *2024 – 2027*

Indira Institute of Technology (Polytechnic), Nanded — Diploma in Computer Engineering | MSBTE | Percentage: 87.49 / 100 *2021 – 2024*

N.E.S. Science College, Nanded — Higher Secondary Certificate (HSC - 12th) | MSBSHSE | Percentage: 94.17 / 100 *2021*

Pratibha Niketan High School, Nanded — Secondary School Certificate (SSC - 10th) | MSBSHSE | Percentage: 93.80 / 100 *2019*

ASSESSMENTS / CERTIFICATIONS

- **Fundamentals of Deep Learning** — Provider: NVIDIA | Key Skills: Neural Networks, Model Training & Evaluation, Image Classification
- **Advanced Course on Green Skills and Artificial Intelligence** — Provider: Edunet Foundation (Skills4Future with AICTE & Shell) | Key Skills: Python, NumPy, Pandas, Machine Learning, Deep Learning, Generative AI Basics
- **Claude 101** — Provider: Anthropic | Key Skills: Prompt Engineering, LLMs, AI Tools & Workflows, Generative AI, NLP
- **Java Programming Masterclass** — Provider: Alison | Key Skills: Java Programming, Object-Oriented Programming (OOP)